

SAHIL NIRANJAN

AI/ML Engineer

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Professional Experience

AI/ML Engineer

January 2026 – April 2026

Bank of New York (BNY)

- Co-developed a **novel LLM/GenAI multi-agent treasury verification system** using **LangChain** and **LangGraph** to automate treasury account validation, orchestrate agent workflows, integrate human-in-the-loop checkpoints, and reduce manual processing time by **35%**.
- Built **RAG pipelines** for compliance and treasury document retrieval using OpenAI embeddings and **ChromaDB**, enabling agents to cross-check account records, payment, verification rules, and internal policy documents across **1,200+** reference files
- Developed scalable **Python data pipelines** to automate feature engineering and model-training data ingestion, improving end-to-end data retrieval latency by **44%** under strict financial compliance and auditability requirements.
- Trained and evaluated **Scikit-learn**, **XGBoost**, and **PyTorch** anomaly-detection models on transaction, account, and verification features, improving treasury exception-detection accuracy by **15%** and prioritizing high-risk cases for analyst review.
- Built and maintained an **end-to-end MLOps pipeline** using **MLflow**, **Docker**, and CI/CD automation to support experiment tracking, model retraining, evaluation, and deployment, reducing engineering maintenance overhead by **25%**.

AI/ML Engineer

January 2022 – August 2024

Vivma Software Inc.

- Built an **XGBoost anomaly-detection model** on compliance-review data for a regulated investment banking client, raising flagged-transaction precision from 0.62 to 0.81 (a **31% gain**) at fixed recall and cutting manual review workload by **22%**.
- Engineered large-scale data pipelines processing **1.5 million records per month** across four teams using Python, SQL, Pandas, and **Apache Spark**, consuming from a **Kafka**-backed event stream for real-time ingestion.
- Designed **feature-transformation logic** that converts raw financial data into ML-ready feature sets through 12 standardized steps, supporting downstream modeling for investment banking and capital markets client engagements.
- Developed a **model-validation pipeline** using **Scikit-learn** and **Weights and Biases** that surfaces label drift and feature leakage, reducing data discrepancies by **38%** and enabling reproducible benchmarking across model iterations.
- Standardized **SQL and dbt feature pipelines** across three teams, cutting processing time from 12 days to 8 days (a **33% reduction**), improving data consistency by 25%, and delivering roughly **\$850K** in annual engineering-hour savings.

Publications and Patents

- Patent IN 202211030731 (Granted):** Real-time 3D pose classification system, 97% accuracy. [Link](#)
- Published:** Emotion classification via multi-modal deep learning, in *IJEAST*, Vol. 6, No. 7. [Link](#)

AI / ML Projects

LOB Predictor: Latency-Optimized ML Inference Engine | C++, ONNX, Scikit-learn, Python | ([GitHub](#)) | ([Demo](#))

- Trained a gradient-boosted classifier on **L2 order-book data** with **18 engineered microstructure features**, exported via **ONNX** and rewrote inference in **C++** to reach **0.7ms p99 latency** on a 1M-event load test, backtested for Sharpe and drawdown.

Customer Behavior Analytics Platform | XGBoost, K-Means, Flask, Streamlit, SQL | ([GitHub](#)) | ([Demo](#))

- Deployed a four-model platform (**K-Means segmentation**, **XGBoost churn prediction at 0.91 AUC-ROC**, pattern detection, item-based recommender) across **50,000+ transactions**, served via **Flask API** and **Streamlit**.

Technical Skills

Programming: Python, R, SAS, C++, SQL

Machine Learning: Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, ONNX, Pandas, NumPy, anomaly detection, classification, feature engineering

LLMs and Deep Learning: Transformers, BERT, Hugging Face, LangChain, LangGraph, OpenAI API, RAG Pipelines, ChromaDB, multi-agent systems

Data Engineering: Apache Spark, Apache Kafka, Airflow, dbt, PostgreSQL, MongoDB, Tableau

MLOps and Deployment: MLflow, Weights & Biases, Docker, Kubernetes, GitHub Actions CI/CD, AWS, FastAPI, Streamlit, model monitoring

Education

Northeastern University <i>Masters of Professional Studies in Analytics</i>	Boston, USA <i>June 2026</i>
International Institute of Information Technology Bangalore <i>Advanced Certificate Program in Data Science</i>	Bangalore, India <i>October 2023</i>
Guru Gobind Singh Indraprastha University (GGSIPU) <i>Bachelor of Technology in Computer Science and Engineering</i>	Delhi, India <i>June 2022</i>